Practice Problems
Calculus Physics II

Name: Key **Date:** _____

Chapter 53 – EM Waves

1) The source of all electromagnetic waves is

A) static charge.
B) nucleus of the atom.
(C) accelerating electric charges.
D) crystalline fluctuations.
E) none of the above

2) Which type of wave can travel through vacuum?

A) Mechanical Waves
B) Sound Waves
(C) Electromagnetic waves
D) Water waves

3) Which of the following is fundamentally different from the others?

(A) sound waves
B) X-rays
C) gamma rays
D) light waves
E) radio waves

4) The time for one cycle of a wave is known as the

(A) Period of the wave.
B) Wavelength of the wave.
C) Amplitude of the wave.
D) Velocity of the wave.
E) Frequency of the wave.

5) The distance a wave travels in one cycle is known as the

A) Period of the wave.
(B) Wavelength of the wave.
C) Amplitude of the wave.
D) Velocity of the wave.
E) Frequency of the wave.

6) The amplitude of a wave is measured from the

A) Peak to peak of the wave.
(B) Equilibrium of the wave.
C) Crest to crest of the wave.

7) The number of cycles per second of the wave is known as the

A) Period of the wave.
B) Wavelength of the wave.
C) Amplitude of the wave.
D) Velocity of the wave.
E) Frequency of the wave.

8) If an electron vibrates up and down 1000 times each second, it generates an electromagnetic wave having a

A) wavelength of 1000 m.
B) wavelength of 1000 km.
C) speed of 1000 m/s.
D) period of 1000 s.
E) frequency of 1000 Hz.

9) The frequency of a wave is 60 Hz and the wavelength is 50 m, what is the velocity of the wave?

$v = f\lambda = 3{,}000\ m/s$

10) The period of a wave is 0.2 seconds, and the wavelength is 3 m, what is the velocity of the wave?

$v = f\lambda \quad | \quad v = \frac{\lambda}{T} = 15\ m/s$

11) The frequency of a microwave signal is 9.76 GHz. What is its wavelength in meters in vacuum?

$v = c = 3\times10^8\ m/s \quad | \quad v = f\lambda = c \quad | \quad \lambda = \frac{c}{f} = 0.03\ m$

12) A certain part of the electromagnetic spectrum ranges from 200 nm to 400 nm. What is the highest frequency associated with this portion of the spectrum? (Assume EM wave is in vacuum.)

highest f → shortest λ

$c = v = 3\times10^8\ m/s \quad | \quad f = \frac{c}{\lambda} = 1.5\times10^{15}\ Hz$

13) If a visible light signal and a radio signal were emitted simultaneously from Alpha Centauri, the first to reach Earth would be the

A) light signal.
B) radio signal.
C) both the same time

14) Which travels with the greatest speed in a vacuum?

A) radio waves
B) light waves
C) X-rays
D) gamma rays
E) all have the same speed

15) If the distance from earth to the Sun is 1.5×10^{11} m, if the Sun were to disappear at this moment, how much time would it take for people on earth to know the sun disappeared?

$v = c = \frac{\Delta x}{\Delta t} \quad | \quad \Delta t = \frac{\Delta x}{c} = 500\,s = 8.3$ minutes

16) If at a particular time it takes 20 minutes for the radio signal to get from NASA on earth to the Mar's rover on mars, how far away is the mars rover from earth at that time?

$v = c = \frac{\Delta x}{\Delta t} \quad | \quad \Delta t = 1200\,s \quad | \quad \Delta x = c\,\Delta t = 3.6 \times 10^{11}$ m

17) Bob is on a planet that is 20 lightyears from the earth and Cathy is on earth, if Bob sends a radio signal wishing Cathy a happy 30th birthday, how old will Cathy be when she gets the signal?

A) 30 years old
B) 20 years old
C) 40 years old
D) 50 years old

30 yrs old + 20 yrs = 50 yrs old

↗ time to get to Cathy

18) Which wavelength of EM waves would have the most energy

A) $\lambda = 100$ nm
B) $\lambda = 1{,}000$ nm
C) $\lambda = 500$ nm
D) $\lambda = 5{,}000$ nm

$E = hf = \frac{hc}{\lambda}$

smaller λ → larger Energy

19) How much energy is in one photon of blue light that has a wavelength of 420 nm?

$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\ J\cdot s)(3 \times 10^{8}\ m/s)}{420 \times 10^{-9}\ m} = 4.7 \times 10^{-19}$ J

20) How many green photons ($\lambda = 530$ nm) are in green light with an energy of 0.01 J?

Energy in one Photon $= \frac{hc}{\lambda} = 3.75 \times 10^{-19}$ J

of Photons $= \frac{E_{Total}}{E_{one}} = 2.7 \times 10^{16}$ photons

21) A source emitting EM waves is moving towards you, how would you perceive the frequency of the EM waves?

A) You would perceive a higher frequency.
B) You would perceive a lower frequency.
C) You would perceive the same frequency.

22) A source emitting EM waves is moving away from you, how would you perceive the frequency of the EM waves?

A) You would perceive a higher frequency.
B) You would perceive a lower frequency.
C) You would perceive the same frequency.

23) If a star is red shifted when viewed from earth, which direction is it moving relative to earth?

A) Towards earth.
B) It is not moving relative to earth.
C) Away from earth.

24) The star ship Enterprise is moving away from earth at 5 x 10^6 m/s, the Enterprise emits a signal towards earth of 8 x 10^{14} Hz, determine the frequency it is detected on Earth. $V = -5 \times 10^6$ m/s

$$f' = f\sqrt{\frac{1+\frac{v}{c}}{1-v/c}} = 7.87 \times 10^{14}\ Hz$$

25) The star ship Enterprise sends a signal to earth when moving relative to earth is 3.4 x 10^{14} Hz, earth receives the signal and measures it to be 3.2 x 10^{14} Hz, determine the velocity of the Enterprise relative to Earth and is it moving towards or away from Earth?

$$V = \left[\frac{\left(\frac{f'}{f}\right)^2 - 1}{\left(\frac{f'}{f}\right)^2 + 1}\right]c = +1.82 \times 10^7\ m/s$$

towards earth

26) A LED is a

A) Light Emitting Diode.
B) Light Enabling Device.
C) Light Emitting Device.
D) Laser Emitting Diode.

27) A Laser is monochromatic, what does this mean about the Laser?

A) It has one battery.
B) It emits one beam.
C) It is collimated.
D) It emits a single frequency.
E) It emits multiple frequencies.

28) LASER stands for

Light Amplification by Stimulated Emission of Radiation

KEY

1) C
2) C
3) A
4) A
5) B
6) B
7) E
8) E
9) 3,000 m/s
10) 15 m/s
11) 0.03 m
12) 1.5×10^{15} Hz
13) C
14) E
15) 500 s
16) 3.6×10^{11} m
17) D
18) A
19) 4.7×10^{-19} J
20) 2.7×10^{16} photons
21) A
22) B
23) C
24) 7.87×10^{14} Hz
25) 1.82×10^{7} m/s
26) A
27) D
28) Light Amplification by Stimulated Emission of Radiation